

itself on channels in the Publish Channel List. This Publish Channel List includes all 20 MHz channels in the 2.4 GHz frequency band and 5 GHz band (if supported) that are allowed per regulation in the geographical location.

The Commissionable device iterates between advertising itself on the Default Publish Channel and on channels in the Publish Channel List per behavior described in [WFA-USD](#). For Commissionable devices to advertise on Dynamic Frequency Selection channels within the 5 GHz band, full compliance with the regulatory requirements of the specific geographical location is mandatory.

The Commissionable device can prioritize advertising on channels in the Publish Channel List in which it determines the presence of an IEEE 802.11 Basic Service Set. The Commissionable device can determine presence of an IEEE 802.11 Basic Service Set on a channel by scanning that channel passively (i.e., SHOULD NOT send Probe Requests) in order to reduce unnecessary transmissions in the shared spectrum.

The detection of an IEEE 802.11 beacon on a particular channel serves as a strong indication of the presence of an IEEE 802.11 Basic Service Set operating on that same channel. Moreover, within the IEEE 802.11 beacon, a Commissionable device can find information about the regulatory domain it is currently located in, specifically in the Country Information Element.

#### 5.4.2.6.3. Publisher Operating Parameters

Commissionable devices SHALL create a Publish instance by invoking the Publish Method, using argument values specified in [Table 74](#), “Publish method arguments values for discovery”. Little-endian encoding is used for all argument values.

Table 74. Publish method arguments values for discovery

| Argument              | Description                                                                                                                                        | Value for discovery                                                                                                                                                                                                                                       |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| service_name          | UTF-8 name string which identifies the service                                                                                                     | _matterc._udp                                                                                                                                                                                                                                             |
| matching_filter_tx    | Ordered sequence of <length,value> pairs to be included in the Publish message                                                                     | NULL                                                                                                                                                                                                                                                      |
| matching_filter_rx    | Ordered sequence of <length,value> pairs that specify further the response conditions beyond the service name used to filter the Subscribe message | NULL                                                                                                                                                                                                                                                      |
| service_specific_info | Sequence of values that are conveyed in the Publish message                                                                                        | Ordered sequence:<br><8-bits, <a href="#">Device OpCode</a> >,<br><16-bits, <a href="#">Device Information</a> >,<br><16-bits, <a href="#">Vendor ID</a> >,<br><16-bits, <a href="#">Product ID</a> >,<br><variable-bits, <a href="#">Extended Data</a> > |